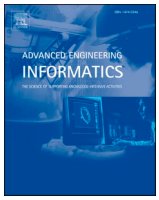
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Full length article



# A status digital twin approach for physically monitoring over-and-under excavation in large tunnels

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ABSTRACT

Accurate assessments of over-and-under-excavation volumes during tunnel construction are critical for stability and structural safety. Over-and-under-excavation assessment is typically time-consuming as it is undertaken manually and is prone to inaccuracies that adversely impact quality, safety, and costs. To overcome this problem, a status digital twin (to monitor physical conditions) is developed to assess over-and-under excavation during tunneling, enabling a robust and automated assessment of volumes of earth to be determined. The development of the status digital twin comprises three stages: (1) creating an as-designed building information model and as-built point cloud model; (2) producing an automated status update with newly added data; and (3) calculating the of over-and-under excavation volume by building a Triangulated Irregular Network (TIN). The status digital twin is tested in a large-scale tunnel project to evaluate its feasibility and effectiveness. The average over-and-under excavation thickness is 0.05 m, and the mean error of volume calculation is 4.54 %. The results demonstrate that the status digital twin can accurately assess over-and-under excavation during the construction of large-scale tunnel projects by providing instant feedback concerning over/under excavation, thus enabling quality to be effectively managed and costs to be controlled.

## 1. Introduction

Drilling and blasting are popular excavation methods in tunnel construction[56]. When drilling and blasting are used, over-and-under excavation (also called overcut and undercut) is likely to occur more often than using other construction methods, such as Tunnel Boring Machines (TBMs)[25]. Over-excavation is defined as an excessively blasted rock area beyond the tunnel's designed contour line. Oppositely, under-excavation is defined as the remaining rock area within the designed contour line in the tunnel. Over-and-under excavation can negatively impact the peripheral rock mass around an excavation area, affecting a tunnel's stability and structural safety and increasing construction costs[22,57]. Monitoring excavation quality is a manually laden process and, as a result, is time-consuming, with estimates of excavated material often inaccurate, adversely impacting project costs [2,22].

While several rule-based or artificial intelligence (AI) methods have been developed to assess over-and-under excavation accurately, it remains a pervasive problem during tunnel construction [2,12,36,13,35,22,43]. With this in mind, this research addresses the following question: *How can over-and-under excavation be accurately assessed during the construction of large-scale tunnel projects*? This research proposes that a status digital twin is used for basic condition monitoring applications to assess and manage the excavation process. Creating operating parameters with visualization tools is core to developing a status digital twin. Thus, this research utilizes a non-invasive three-dimensional (3D) point cloud-based method to accurately and automatically assess a tunnel's over-and-under excavation during construction.

Thus, the paper commences with a brief review of the three-dimensional (3D) point cloud and semantic segmentation application in tunnel construction (Section 2). Details about the proposed status

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digital twin, which utilizes a point cloud and Triangulated Irregular Network (TIN), are then presented (Section 3). A case study is then used to test and validate the digital twin in a real-life project (Section 4). A discussion about the contributions and limitations of the developed status digital twin is presented (Section 5) before the paper's conclusions (Section 6).

## 2. Background to the research

Digital twins can take various forms depending on their function [31,41]. As mentioned above, the development of a status digital twin forms the focus of this paper. However, a cursory glance at the literature reveals two other types of digital receiving attention: (1) *Operation twins*, which are used to adjust operating parameters based on linked actions and/or workflows; and (2) *Simulation twins*, which are used to predict how an objective or device responds to operational conditions in the future – this is key for predictive maintenance and product development.

With advances in AI and sensing technologies coming to the fore, digital twinning has become a popular research topic, as evident by the sheer number of research papers dealing with this subject matter (e.g., [7,24,8,37,9,27,42]. Undertaking a detailed review of digital twin literature is outside the scope of this paper as it focuses on developing a new approach for assessing over-and-under excavation during tunnel construction. However, within the context of tunnel construction, readers are advised to refer to the works of Li et al. [28] and Li et al. [29].

### 2.1. Three-dimensional (3D) point cloud for tunnel construction

Due to the accuracy and efficiency of 3D point clouds, they have been widely used in construction for numerous applications, such as progress monitoring, building performance examination, safety inspection, and retrospective building information models (BIM) from 'as-builts' documentation[32,38,49,54]. Several studies have also applied point clouds in tunnel construction for cross-section extraction[3,10], deformation measurement[18,48], and reconstruction[11,55]. However, using point clouds to assess over-and-under excavation in tunnel projects has been limited to the work of Huang et al. [22], who employed image-matching and cubic parametric interpolation spline methods to visualize and calculate the volume of over-and-under excavation.

While Huang et al.'s [22] work enabled significant headway to be made with the calculation of over-and-under excavation, improvements are required to their method to further improve its accuracy by: (1) automatically segmenting part of a tunnel from a point cloud; and (2) the process to determine geometric calculations. Point cloud semantic segmentation is the task of clustering parts of points that belong to the same object class. Studies examining point cloud semantic segmentation can be categorized as: (1) model-driven approaches; and (2) data-driven approaches.

### 2.2. Point cloud semantic segmentation

Model-driven approaches classify point clouds into different segments based on predefined labels (e.g., descriptions of objects of interest). In the context of point cloud semantic segmentation, a plethora of model-driven approaches were developed. For example, the model fitting-based approaches aim to fit shapes onto point cloud data to detect objects' geometric primitives. Two of the most widely used model fitting-based approaches are the Hough Transform (HT)[5]and Random Sample Consensus (RCS)[16]. Still, they require prior knowledge of predefined shapes or surfaces[14]. However, the RCS has been proven to perform better in terms of efficiency and robustness than the HT[1,15].

Data-driven methods predict objects using training data[44,51,52]. For example, the feature clustering-based method aggregates points with similar attributes into clusters[6,58]. Implementing this method initially requires extracting predefined features from the points (e.g., normal surface vector, height difference, and saliency). Although feature clustering-based methods are proven to perform better than region-growing methods[1,45], they also suffer from noise or outliers, and designing a practical feature is a critical factor for their successful implementation[15].

With significant advancements in AI, deep learning-based data-driven approaches have been widely used for segmenting point clouds [17,50]. Compared with the approaches mentioned above, a deep learning algorithm automatically learns predictive features from many labeled samples, avoiding the need for expert-defined features [23,26,40]. However, semantic segmentation of point clouds remains challenging due to their irregular format and varying point densities [33].

### 2.3. Volume calculation with triangulated irregular network

The TIN has been widely used as a geometric processing tool in various manufacturing and engineering fields. It is a versatile tool representing the spatial distribution of physical elements measured in discrete locations[46]. The TIN differs from point clouds by offering a continuous surface to represent geometric data, which allows different interpolation techniques to be applied for finer value extraction or computation. For example, the vertices of a TIN can be used to calculate the slope and aspect[21], gravity-specific curvature, and other land surface parameters[34].

The TIN has been combined with various sensing techniques for volume measurement. For example, large TINs are constructed to assess the volume of earthwork[19]) and lake water from remote-sensing data, and a smaller range of TINs can also be used to measure the crown projection area and volume of trees[30]. Combined with dedicated algorithms, TINs can be generated from different data sources and serve as a basis for more advanced geometric calculation tasks. Using TINs for tunnel excavation volume calculation has received limited attention. To overcome the challenges as mentioned earlier associated with automated point cloud segmentation and excavation volume calculation, two issues require consideration: (1) new end-to-end deep learning algorithms need to be developed to directly extract and learn point cloud features and produce segmented results without data representation transformation[39]; and (2) the use of the TIN-based method to calculate the volume of over-and-under excavation.

## 3. Status digital twin of excavation

To recap, the aim is to develop an approach to evaluate over-and-under excavation in large-scale tunnel projects accurately. The proposed approach consists of three significant steps: (1) initialization of the status digital twin consisting of an as-designed BIM; (2) automated periodic status update with newly added data; and (3) the calculation of over-and-under excavation volume using a TIN-based algorithm. The framework of the status digital twin is presented in Fig. 1. Notably, the digital twin is continuously updated with automated steps each time a new scan is registered.

### 3.1. Initialization with as-designed building information model

Autodesk Revit® and Dynamo generated the as-designed building information model (BIM) based on computer-aided design (CAD) drawings. Dynamo is a piece of visual programming software that runs on the Revit® platform for parametric design. Based on the CAD files, Dynamo reads the parameters in the drawings, generates surfaces after connecting control points, and finally fuses to form a 3D tunnel model. The workflow of the 'as-designed' BIM is presented in Fig. 2.

### 3.2. Automated status update with point clouds

Pre-processing the laser-scanned point cloud data is an important

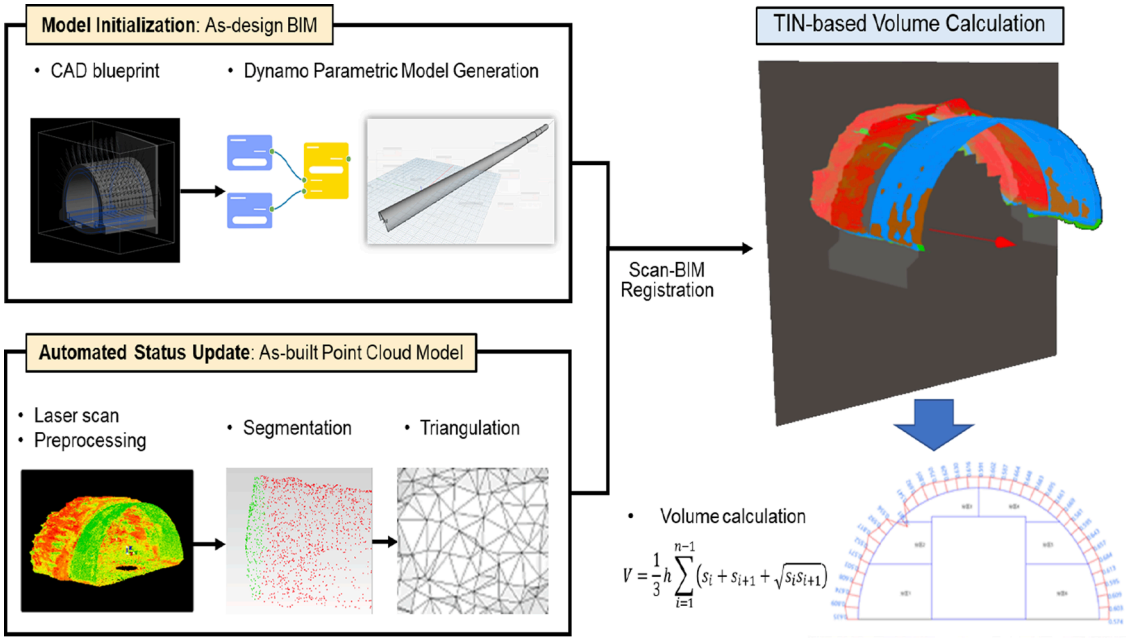
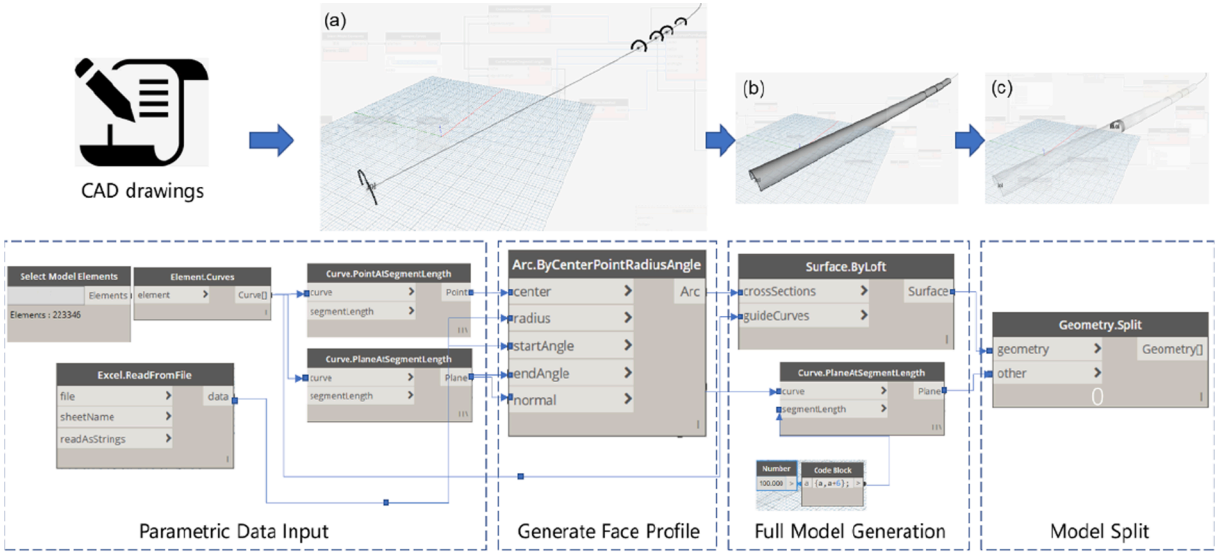



**Fig. 1.** The framework of the status digital twin model.




(a) Define the central line and tunnel face contours; (b) Generate the tunnel's full-mesh model; (c) Extract a tunnel mesh ring at a given location.

**Fig. 2.** The workflow to generate the 'as-designed' BIM using Dynamo.

step for the as-built 3D reconstruction process; it is necessary to ensure semantic segmentation's accuracy. Thus, point cloud data pre-processing techniques are adopted. The procedure to pre-process the point cloud data is described in the following four steps.

1. *Compression of Tunnel Face Point Cloud*: Fine-grained point cloud data generated by laser scanners is usually too heavy to be processed directly. An octree-based approach was adopted[20]to compress the point cloud to preserve key point cloud information and simultaneously reduce the number of points (for example, by a ratio of 50 %)
2. *Point Cloud Filter*: After raw point cloud data compression based on the normal vector, the Random Sample Consensus (RANSAC) method was adopted in combination with space decomposition to simplify the input point cloud further while keeping minimum shape loss. The space is partitioned into voxels, each containing a subset of the entire point cloud. Moreover, the RANSAC can help remove outliers from the target surface. The side length of the voxel and the minimum number of points inside a voxel are key parameters for space decomposition, and the former is proportional to the point cloud data simplification rate
3. *Automated Segmentation of Tunnel Face from Point Cloud*: Inside a tunnel, the tunnel face and surrounding rock are closely connected with homogenous texture, making it difficult to separate by generic point cloud segmentation methods. To avoid manual corrections after heuristic segmentation methods, an encoder-decoder network architecture adapted from the PointNet++ [40] was adopted to achieve accurate point cloud segmentation, taking advantage of the shape before the tunnel cross-sections. This state-of-the-art point cloud segmentation network can extract features with different granularities using a single network. The network employs a

hierarchical structure that recursively applies PointNet on a nested partitioning of the input point set to capture features of the whole point cloud and point subsets at different scales. Since the classification part of PointNet++ is designed for the entire point cloud, it is removed from the pipeline, and only the segmentation part is kept for further training. The encoder part follows a down-sampling process. Multi-level down-sampling is adopted through multiple set abstraction structures to obtain point-wise features of different sizes. The final set abstraction output corresponds to global features for the whole point cloud. The essential component set abstraction comprises three modules: (1) sampling; (2) grouping; and (3) PointNet. The decoder part for the segmentation task follows an up-sampling process. The point-wise feature is obtained by reverse interpolation and skip connection, which contains local and global information. The propagated features of the original set of points are computed using the following equations

$$f^{(j)}(\mathrm{x}) = \frac{\sum_{i=1}^{k} w_i(x) \bullet f_i^{(j)}}{\sum_{i=1}^{k} w_i(x)} \tag{1}$$

$$w_i(x) = \frac{1}{\mathrm{d}(\mathrm{x}, x_i)^p}, j = 1, \cdots, C \tag{2}$$

where $f^{(j)}(\mathrm{x})$ corresponds to the feature vector, $w_i(x)$ The interpolation weight depends on the inverse distance weighted average based on $k$ nearest neighbors. Features for each point can be obtained with this feature propagation rule. The encoder-decoder network architecture is illustrated in Fig. 3.

4. *Triangulation and Registration*: After removing the outliers, the next step is to generate a Delaunay triangulation mesh from sample points using the existing library functions in MatLab as per Qhull[4]. Scan-BIM registration aims to align the scanned point cloud data with the 'as-designed' building information model. To achieve higher alignment accuracy, the tunnel's scanned point cloud data are used to calculate the transformation matrix between the 'as-designed' BIM. The 'as-designed' BIM is processed using a standard geodetic coordinate system from design drawings. The coordinate system of the 3D point cloud model is the laser scanner's internal coordinate system. Calculating the deviation between the 'as-designed' and 'as-built' BIM models in the same coordinate space determines the quantity of over-and-under excavation

### 3.3. Volume estimation with TIN

The final step is to analyze the over-and-under excavation. Fig. 4 presents the workflow of the TIN-based method for over-and-under excavation volume calculation and comprises three sub-steps:

1. *Clipping and Plane Intersection:* The 'as-built' and the 'as-designed' BIM are 3D mesh models (TINs), which comprise many triangular planes with three vertices, and each vertex has a unique coordinate value. The calculation commences by defining a clipping plane that slides along the central line of the tunnel with a fixed interval. Each time the clipping plane is positioned, it will intersect with the 'as-built' and 'as-designed' models, and the intersection points are connected to become two contours, respectively. These contours are then used to form a cross-sectional slice representing the extent of over and undercut.
2. *Cross-sectional Thickness and Area Calculation:* Given a cross-sectional slice, point-plane distance equations can calculate the exact excavation thickness distributed along the contours. The thickness at a given location is represented by picking the vertex on the contour from the scanned model and calculating the distance from that vertex to the as-designed TIN. Given a plane $Ax+By+Cz+D=0$ (a triangle in the as-designed TIN), the distance ($d$) from a point outside the plane $(x_0, y_o, z_0)$ to the plane, is as noted in (3):

$$d = \frac{Ax_0 + By_0 + Cz_0 + D}{\sqrt{A^{2+}B^{2+}C^2}} \tag{3}$$

The distance (d) represents how far the scanned surface is from the designed one (i.e., the excavation thickness). The area of the cross-sectional slice can then be computed by summing the area of triangles inside the sliced plane.

3. *Volume calculation:* The sampling calculation method is used to derive the cross-sectional area from slice *1* to slice *n*, and the column volume calculation method (see (4)) is employed to approximate the overall volume of over/undercut

$$V = \frac{1}{3}h\sum_{i=1}^{n-1}(s_i + s_{i+1} + \sqrt{s_i s_{i+1}}) \tag{4}$$

where $V$ is the overall volume of over and undercuts in the drilling and blasting cycle from slice *1* to *n*; $h$ is the sampling interval in the direction of drilling and blasting ($h$ is generally 0.1 m); $s_i$ represents the cross-sectional area of slice $i$.

In the next section of this paper, the developed status digital is tested

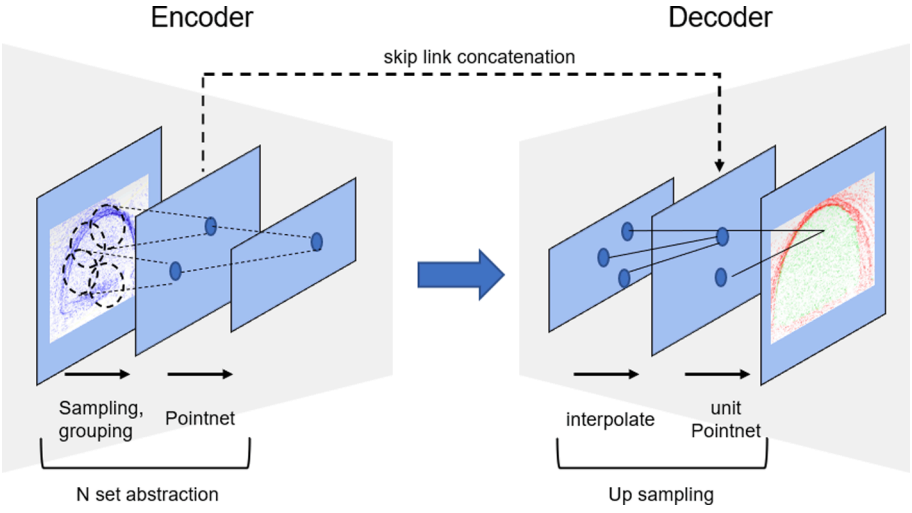



**Fig. 3.** The simplified network architecture of PointNet++ for the tunnels' point cloud segmentation.

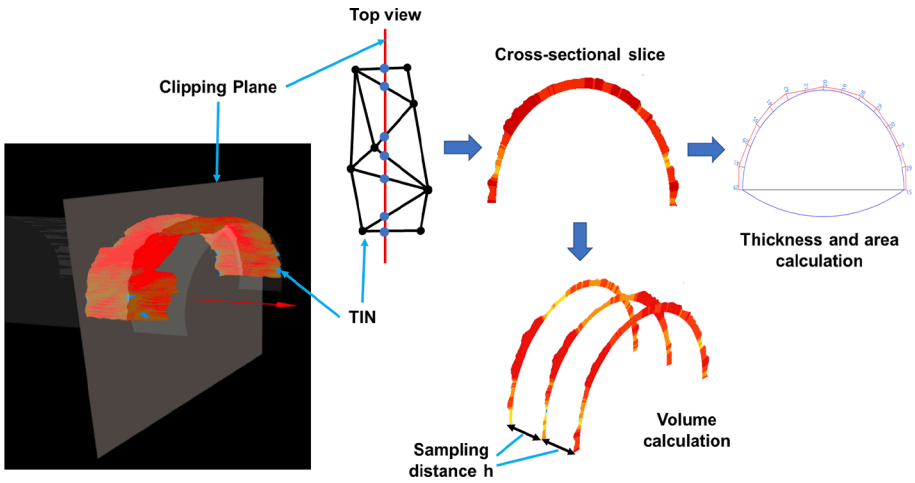



**Fig. 4.** Workflow of TIN-based over/undercut analyses.

and validated in a real-life tunnel construction project.

## 4. Case study

The expressway project in Guizhou, China, is used to test the feasibility of the proposed status digital twin. As shown in Fig. 5, the project starts in the east of Leishan County (Guizhou, China), passes through a series of neighboring towns and villages, and ends at Zhongcheng in the north of Rongjiang County (Guizhou, China). The total length of the expressway project is 72.45 km, with an estimated budget of 12.29 billion yuan (approx. US$ 1.84 billion). The project has a total of 18 tunnels, with four of them longer than three kilometers.

Fig. 5 shows the distribution of the four long tunnels, which share the same design pattern. The geological structure of the tunnels is complex, mostly highly weathered sandstone or slate, where the boundary of the rock formation and the local section of the occurrence change will lead to more overcut and undercut.

Fig. 6 presents the rock quality distribution for the Leigongshan Tunnel, the longest among the 18 tunnels in this project. The tunnel has a total length of 4960 m with host rock quality between level IV and V according to the Code for Design of Road Tunnel (JTG D70-2004) in China (quality levels range from I to VI, a smaller number indicates better rock quality), where the uniaxial compressive strength of rocks changes between 8 and 25 MPa.

### 4.1. Data collection

In this project, two types of data are collected: (1) tunnel point cloud data using laser scanners; and (2) coordinates of point cloud obtained using a Real-Time Kinematic (RTK) device. As shown in Fig. 7, the point cloud data was acquired using a Terrestrial Laser Scanning (TLS) device (i.e., Leica ScanStation P40). Point cloud samples for the entire tunnel

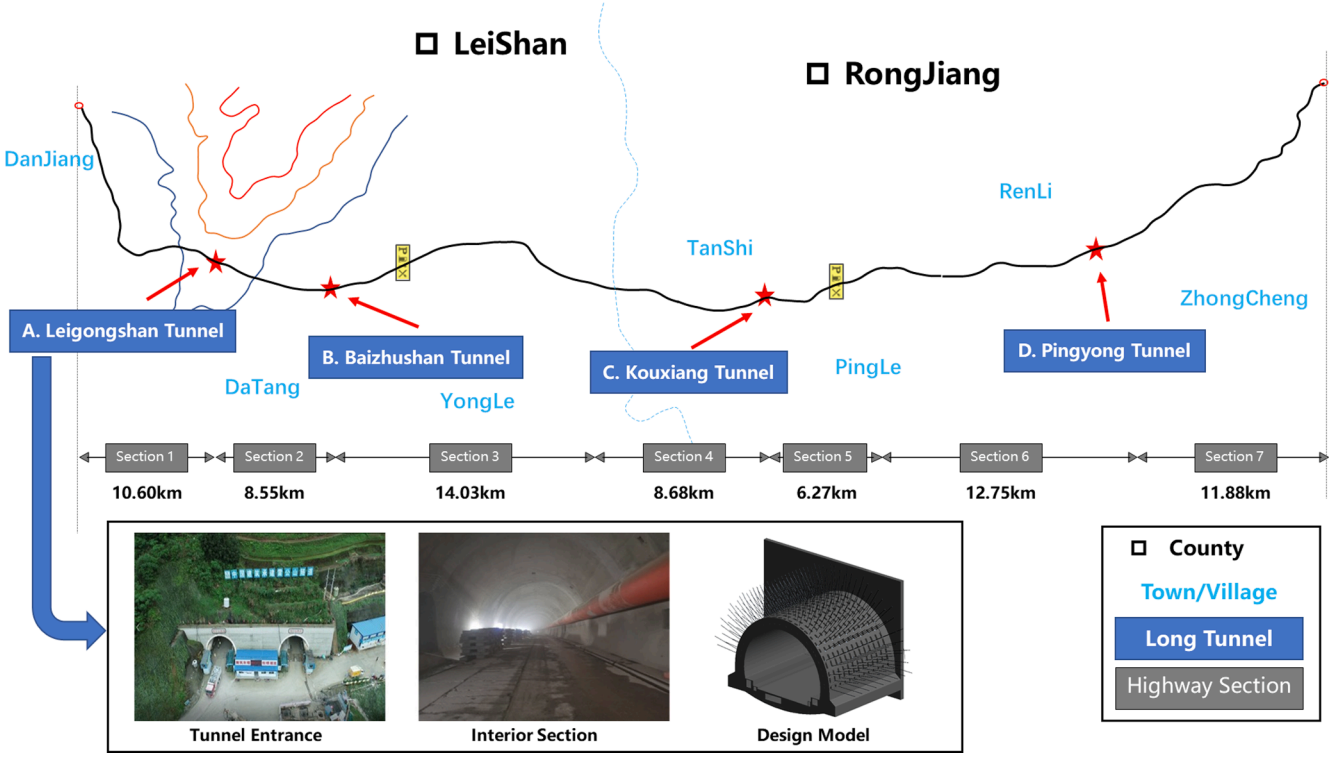



**Fig. 5.** Map of Leirong expressway project.

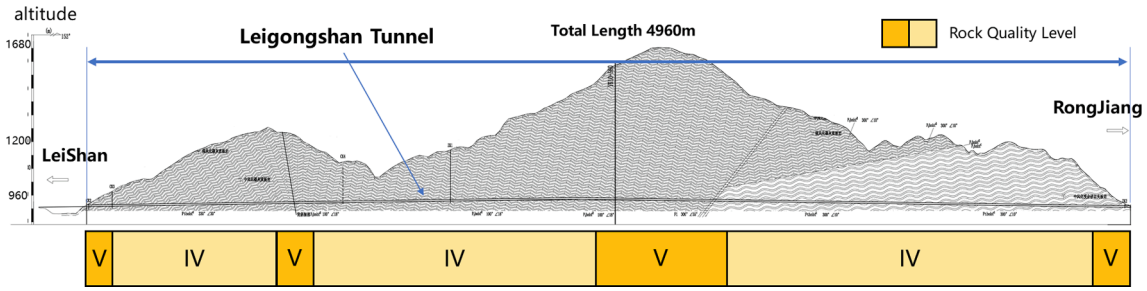



**Fig. 6.** Rock quality distribution of Leigongshan Tunnel.

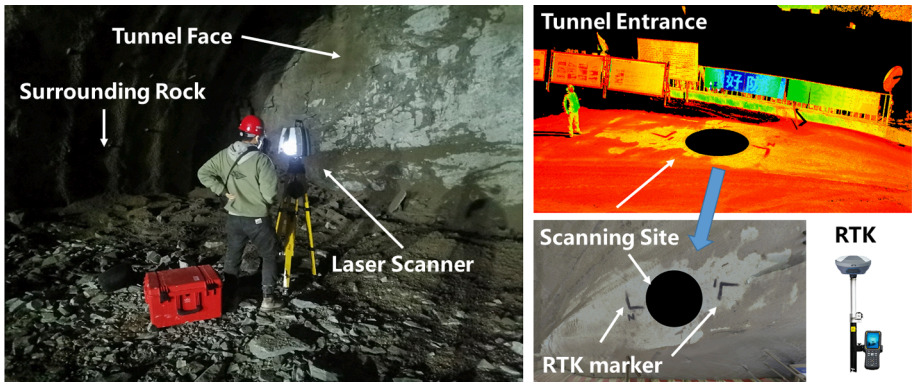



**Fig. 7.** Data collection steps in front of the tunnel face and entrance.

were made approximately every 40 m starting from the tunnel entrance, and all the scanned samples were concatenated until reaching the tunnel face. "L" shaped red marks were made at the tunnel entrance to provide visual cues for coordinates measurements with the RTK station. The measured coordinates (in the WGS-84 coordinate system) were assigned to the point clouds to enable the alignment of the 'as-designed' BIM and 'as-built' point clouds.

### 4.2. Data pre-processing

Dedicated software (i.e., Cyclone) was first used to pre-process the obtained tunnel face point cloud data to form a preliminary processed *xyz* file, which provided a basis for subsequent gridding to process the original scanned point cloud.

Automatic point cloud segmentation was achieved using a customized PointNet++ algorithm by removing the classification module from the pipeline. The network was trained with 21-point cloud samples and validated and tested with 6 and 3 samples, respectively. Each point cloud sample initially had over one million points and was down-sampled to 20,480 points to save memory. The distribution of points between the tunnel face and tunnel surrounding rock classes was about 15 % and 85 %.

The network was trained with a learning rate of 0.00025 for 60 rounds, which resulted in an average test accuracy of 0.94 and mIoU (mean Intersection over Union) of 0.91. This segmentation network can help separate the surrounding rock from the tunnel face and provide input for mesh conversion. Fig. 8 presents examples of automatic point cloud segmentation.

Next, mesh processing software Geomagic Wrap 2017 was used to achieve further point cloud mesh conversion. The post-processed point cloud model was meshed and repaired to form a complete and smooth mesh model. Finally, 3D mesh models of the tunnel face and surrounding rocks with high geometric precision are obtained, as noted in Fig. 9.

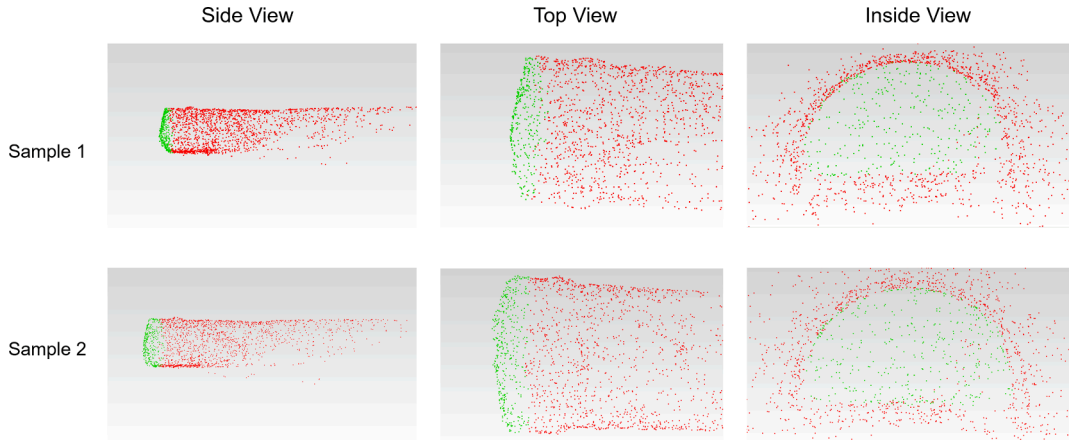
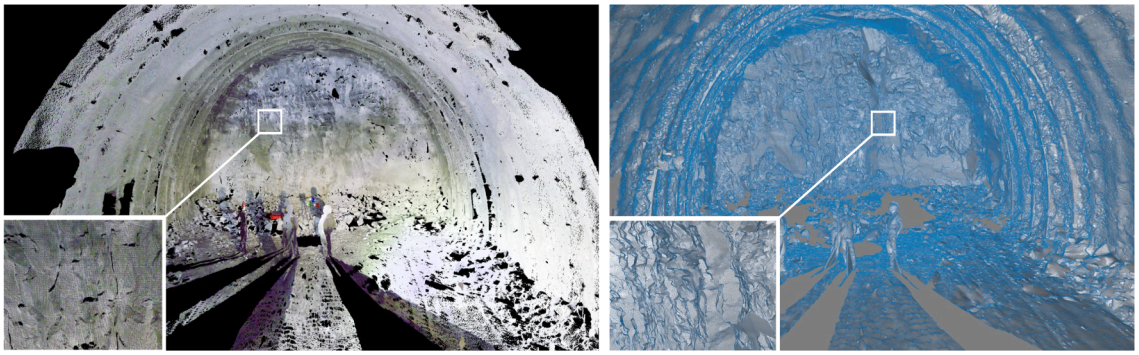



**Fig. 8.** Examples of automatic point cloud segmentation.



(a) Point cloud of tunnel face with surrounding rock; (b) Scanned model after being meshed.

**Fig. 9.** Examples of as-built 3D models of a tunnel.

### 4.3. Results and analyses

Three long tunnels were selected for evaluation, and 30 scanned samples were collected and compared against manual excavation recordings. The tunnel name, number of samples, level of host rock, the thickness of C20 injection, and upper step arc length are shown in Table 1. The pouring volume of shotcrete in all cases in selected tunnels was obtained from field records and used as ground truth for over/under excavation volume computation (the pouring volume can be different from the final volume after solidification but was the only available reference data).

The TIN-based volume estimation method was compared with a manual measurement solution. The manual solution works as follows: the surveyor chooses a random location along the upper step arc inside a random cross-sectional plan and then notes the distance between the design surface (indicated by a total station) and the actual surface as excavation thickness d. Then, the estimated excavation volume can be calculated by (5):

$$V = d^{*}l^{*}h \tag{5}$$

where $V$ is the volume of over and undercuts in one drilling and blasting cycle; $h$ is the length in the direction of drilling and blasting; l represents the upper step arc length. The manual measurement method is often used as a standard solution for its simplicity. Four sets of average over-and-under excavation thickness and four sets of manually calculated volumes were collected.

To verify the effectiveness of the TIN-based volume estimation method, it was compared with the manual measurement solution using the actual pouring volume as a reference. The results of the comparisons are shown in Table 1 and Fig. 9.

From Fig. 10(a), the error of volume calculation of the TIN-based approach and manual measurement was 4.54 % and 25.72 %, respectively (compared to the actual pouring volume). The error of volume calculation in groups 1, 3, and 4 were all less than 20 %, which meets the requirements of tunnel construction inspection. However, in group 2, the manual calculation error cannot fit the requirements. The calculation error of using our proposed approach is significantly smaller than the manual measurement error. Fig. 10 (b) shows that the thickness of the average over-under-excavation using the status digital twin approach and manual measurement was 0.05 m and 0.09 m, respectively. The deviation of the average thickness for over-and-under excavation using the status digital twin was smaller than that of the manual measurement under multiple sections.

**Table 1**
Parameters of selected tunnel excavation samples from three tunnels.

| Group ID | Tunnel name | Stake number | Rock grade | C20 injection thickness (cm) | Upper step arc length (mm) |
|---|---|---|---|---|---|
| 1 | Leigongshan | ZK11 + 551.7 ~ 555.2 *m* | IVb | 18 | 19,110 |
| 2 | Taojiang | ZK14 + 74.8 ~ 78.8 *m* | Vc | 24 | 19,370 |
| 3 | Taojiang | ZK14 + 72.4 ~ 74.8 *m* | Vc | 24 | 19,370 |
| 4 | Baizhushan | YK18 + 233.5 ~ 237.1 | IVb | 18 | 19,110 |

## 5. Discussion

Digital twins serve as a transformative bridge between the physical and digital realms, enabling the acquisition of real-time data concerning physical objects' status, working conditions, and position. Despite their potential advantages across the asset life cycle, quantifying specific benefits and business value during construction remains a persistent challenge[31]. This research has developed a status digital twin capable of accurately determining excavation volumes during the excavation process. While immediate quantification of benefits may not be provided in this research, it is crucial to recognize and explore additional tangible advantages, including enhanced safety, increased productivity, and reduced costs.

The proposed status digital twin significantly enhances informed decision-making processes by leveraging its technical attributes, such as data integration and semantic data description. Data integration ensures decision-makers access current information about the physical state, facilitating proactive decision-making. Furthermore, semantic data description improves data interpretability, enabling decision-makers to comprehend the context and relationships among different variables. For example, the attributes derived from the status digital twin primarily involve excavation conditions at various locations, specifically the distribution of over- and under-excavation along contour lines. Such information can assist on-site managers in intuitively understanding excavation quality in different tunnel zones. Additionally, it can aid in establishing the connection between excavation parameters and the volume of the excavation, optimizing parameters for subsequent excavation rounds.

### 5.1. Contributions

The contributions of this research are twofold. Firstly, the volume of the excavation can be calculated automatically to determine whether it is over or under-excavated. The feasibility and effectiveness of this approach are well-suited for application on-site. The developed TIN-based excavation volume calculation method employed a fine-grained computation logic and was validated using the actual pouring volume

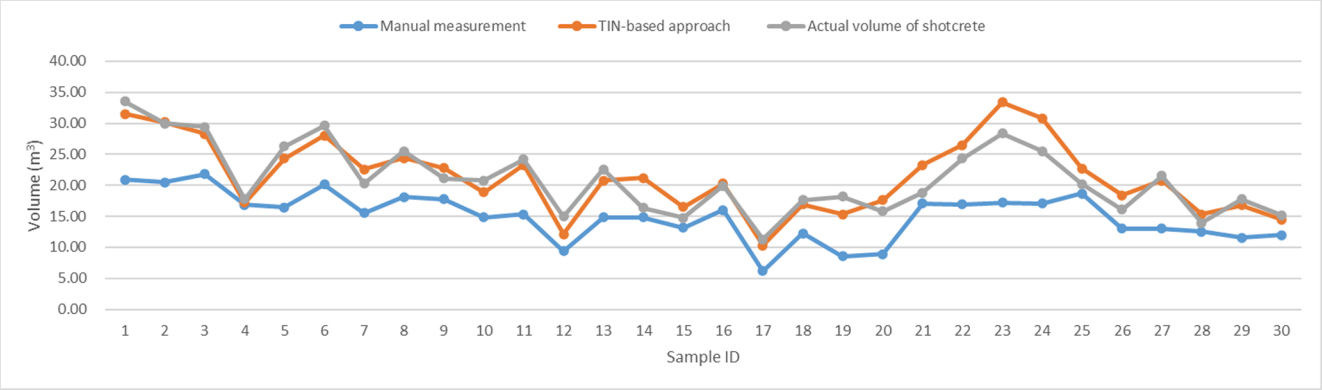



**Fig. 10.** Comparisons between the TIN-based approach and manual measurements.

as ground truth. Previous studies determining over-and-under excavation have only compared their calculations to manual measurements [22]. The comparisons presented in Fig. 10 demonstrated that the TIN-based method provides more stable and accurate volume estimations at the cost of building a point cloud model while considering multiple cross-sectional planes for calculation. Manual measurements exhibited large variations and tended to underestimate the excavation volume. This issue could be attributed to the random selection of the cross-sectional plane for measurement and the variations in working methods among site surveyors. Secondly, a deep learning approach was adopted to automatically segment the tunnel face and surrounding rock based on PointNet ++. Due to the continuity of the rock mass itself and the weak texture characteristics of the tunnel's rock wall, this part's point cloud segmentation is often conducted manually using dedicated software. This automated point cloud processing approach can improve the accuracy and efficiency of status updates.

### 5.2. Limitations

Despite the novelty of the research that has been presented, it needs to be acknowledged that several limitations exist. Ensuring the quality of point cloud data (e.g., 'noise') is a challenge. In the case of a tunnel project, noise in the 'Light Detection And Ranging' (LiDAR)-generated point clouds can materialize from unexpected objects or materials. Thus, the 'noise' in the point cloud renders it difficult to represent the tunnel perfectly after triangulation. New algorithms are needed to address this limitation and improve the point cloud quality[47].

We acknowledge that our proposed status digital twin approach has been validated only in limited tunnel projects and does not apply to all geological conditions. For example, under conditions of abundant water, the mechanical parameters, such as the strength of the rock, will significantly decrease after it is encountered, and the deformation of the surrounding rock will increase, posing several challenges for the on-site application of this method[53]. Furthermore, it should be acknowledged that the dataset used in this research was imbalanced, negatively impacting the model's ability to generalize well to new scenarios and unseen data. To address this limitation, there is a need to develop resampling techniques (i.e., undersampling and oversampling) and new algorithmic approaches (i.e., ensemble approaches). Moreover, the deformation of the surrounding rock will increase with the exposure time of the excavation face after excavation. In this instance, the accuracy of the data obtained during on-site data collection is greatly affected by the timing of on-site measurement collection. To address this issue, we suggest that the frequency of data collection needs to be increased.

To sum up, the research focuses on introducing a new method for assessing the volume of excavation. This method involves a web-based application of a status digital twin, emphasizing functionality rather than user interface, accessibility, and mechanisms for ensuring data accuracy and timeliness. It should be acknowledged that the function (i.e., user interface, accessibility) is significant for digital twins, which are at the core of *operation twins* and *simulation twins*. Therefore, future research will shift its focus towards developing *operation twins* with real-time control and management functionalities and creating *simulation twins* for predictive analysis and scenarios.

## 6. Conclusion

This paper proposed a 'status' digital twin to automatically assess a tunnel project's over-and-under excavation with its development comprising three steps: (1) initialization of the 'status' digital twin consisting of an as-designed BIM; (2) automated status update with as-built point cloud data; and (3) the calculation of over-and-under excavation volume by a TIN-based geometric method. A large-scale tunnel project in Guizhou, China, was used to validate its effectiveness and feasibility.

The results demonstrated that the status digital twin accurately evaluated over-and-under excavation with an average thickness of 0.05 m and an error of volume calculation of 4.54 %, respectively. In this case, site managers, for example, can take action to control the excavation process. Evaluating overcut and undercut during excavation can result in effective managerial interventions to ensure a tunnel's stability and structural safety.

For future research, the status digital twin model can be extended to simulate the excavation process and predict the potential risk caused by under and over-excavating. In addition, further research is required to consider tunnel face conditions (e.g., distribution of cracks and rock layers on tunnel face). The distribution of cracks and rock layers on the tunnel face can lead to different blast-hole distribution and further control parameter variations, which enables more accurate over-and-under excavation control.

## CRediT authorship contribution statement

**Weili Fang:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Weiya Chen:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Peter E.D. Love:** Writing – review & editing, Writing – original draft, Conceptualization. **Hanbin Luo:** Writing – review & editing, Writing – original draft, Conceptualization. **Haiming Zhu:** Conceptualization. **Jiajing Liu:** Conceptualization, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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